

DATASCI 530 – Assignment 3

For this assignment, you will interact with an API and extract information encoded in JSON format. OpenAlex is a fully open catalog of the global research system, with millions of records from different types of sources. You can read more about it here:

<https://docs.openalex.org>

<https://docs.openalex.org/quickstart-tutorial>

Use these tutorials to answer the following questions:

1. How many works have “artificial intelligence” in their titles?
2. Restrict the analysis to works published since 2000 and group by the country of the authors’ affiliations. Produce a table that sorts the countries by the number of publications about artificial intelligence.
3. What are the titles of the 10 most highly cited papers on artificial intelligence since 2000? How many citations do they have? Who are the authors? What institutions are they affiliated with?
4. How has the number of publications with the phrase “artificial intelligence” changed year-to-year since 2000? How does this compare to the use of the terms “LLM” and “GPT”? Repeat the year-to-year analysis for another keyword of your choice. Plot your results in a graph.

New insight questions:

5. Compute an additional table or figure not included in the questions above.
6. What do we learn from this new table or figure?

For this assignment, you will need to use the Works API, which allows you to extract information about individual publications and search by topic, country code, etc. Read the page about how to filter works: <https://docs.openalex.org/api-entities/works>

OpenAlex has a series of Python tutorials on how to interact with the API. Use these as inspiration for your own analysis. The tutorial on Japanese publications, while focusing on a different set of questions, could be a useful reference. See more details here:

<https://docs.openalex.org/how-to-use-the-api/api-overview>

<https://docs.openalex.org/additional-help/tutorials>

For this homework, you are expected to process JSON-formatted data and automate your API calls using control flow.

Deliverable: Submit an HTML file with your results on Canvas. Include Markdown text chunks to write comments on your findings and explain your methodology. You are expected to have

a title for your project and a short section at the beginning summarizing the findings. We recommend that you do not print raw data and only print final tables and/or figures.

When submitting your work, imagine that you are sharing your report with colleagues who have asked you to explain and justify the key decisions and findings from your analysis. The file should be professionally presented, self-explanatory, and have clearly labeled sections.

Rubric:

You will be graded both on the code and the writing using the following point system:

Code - Code runs without issues. - Code is appropriate for answering the questions. - File is appropriately organized, with different sections and comments clarifying the definition of variables and dataset names.	2 points
Description of methodology - Key coding decisions are explained via Markdown chunks. - Choices about how to clean the data are explained. - Quality checks are conducted, such as verifying the sample size of merged datasets, checking for the presence/absence of missing values for key variables, and/or computing any relevant descriptive statistics.	2 points
Results and Visualization - Includes a short summary of the findings. - Results are correctly interpreted and easy to follow. - Includes either tables or graphs to display results from different groups. - All graphs look professional (have axis and title labels, a suitable color scheme, appropriate font sizes, a legend when needed, etc.). - Tables are properly formatted (e.g., have appropriately rounded values, relabeled columns, captions, and are not too long).	4 points
New Insight - Analysis is distinct from the questions already provided. - Clear interpretation of the new analysis.	2 points